



Introduction to CFD (4RC30)

Unsteady flow problems

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Summary of last lecture

- **convection** = **diffusion** + (**source** – **sink**)

$$\text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- **General form of discretized equations:**

$$a_P\phi_P = \sum a_{nb}\phi_{nb} + S_u$$

$$a_P = \sum a_{nb} + \Delta F - S_P$$

- **Spatial discretization schemes: CDS, UDS and Hybrid**
- **Momentum equations: SIMPLE algorithm**

Content of this lecture (Chapter 8)

- Unsteady heat conduction
- Time discretization schemes
- Unsteady convection-diffusion
- Transient SIMPLE
- Example: Vortex Shedding

Unsteady flows

- Unsteady transport equation:

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad} \phi) + S_\phi$$

- Integrated form:

$$\int_{CV} \int_t^{t+\Delta t} \frac{\partial}{\partial t}(\rho\phi) dt dV + \int_t^{t+\Delta t} \int_A \mathbf{n} \cdot (\rho\phi\mathbf{u}) dA dt = \int_t^{t+\Delta t} \int_A \mathbf{n} \cdot (\Gamma \text{grad} \phi) dA dt + \int_t^{t+\Delta t} \int_{CV} S_\phi dV dt$$

Unsteady heat conduction

Unsteady 1D heat conduction: Specific heat [J/(kg K)]

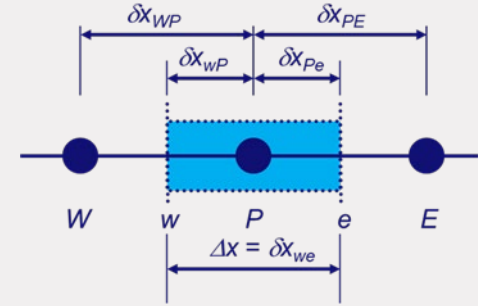
$$\rho C_P \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) + S$$

Thermal conductivity [W/(m K)]

$$\int_{CV} \int_t^{t+\Delta t} \rho C_P \frac{\partial T}{\partial t} dt dV = \int_t^{t+\Delta t} \int_A \mathbf{n} \cdot \left(k \frac{\partial T}{\partial x} \right) dA dt + \int_t^{t+\Delta t} \int_{CV} S dV dt$$

$$\rho C_P (T_P - T_P^0) \Delta V = \int_t^{t+\Delta t} \left[k_e A_e \frac{T_E - T_P}{\delta x_{PE}} - k_w A_w \frac{T_P - T_W}{\delta x_{WP}} \right] dt + \int_t^{t+\Delta t} \bar{S} \Delta V dt$$

Central differencing



Treatment temperature in time

$$\rho C_P (T_P - T_P^0) \Delta V = \int_t^{t+\Delta t} \left[k_e A_e \frac{T_E - T_P}{\delta x_{PE}} - k_w A_w \frac{T_P - T_W}{\delta x_{WP}} \right] dt + \int_t^{t+\Delta t} \bar{S} \Delta V dt$$

Temperatures can be evaluated at different time:

$$\int_t^{t+\Delta t} T_P dt = T_P^0 \Delta t$$

T at old time level t only

$$\int_t^{t+\Delta t} T_P dt = \frac{1}{2} (T_P + T_P^0) \Delta t$$

T at t and $t+\Delta t$

$$\int_t^{t+\Delta t} T_P dt = T_P \Delta t$$

T at new time level $t+\Delta t$ only

Treatment temperature in time

$$\rho C_P (T_P - T_P^0) \Delta V = \int_t^{t+\Delta t} \left[k_e A_e \frac{T_E - T_P}{\delta x_{PE}} - k_w A_w \frac{T_P - T_W}{\delta x_{WP}} \right] dt + \int_t^{t+\Delta t} \bar{S} \Delta V dt$$

Divided by $A\Delta t$ to obtain:

$$\rho C_P \frac{(T_P - T_P^0)}{\Delta t} \Delta x = \theta \left[k_e \frac{T_E - T_P}{\delta x_{PE}} - k_w \frac{T_P - T_W}{\delta x_{WP}} \right] + (1 - \theta) \left[k_e \frac{T_E^0 - T_P^0}{\delta x_{PE}} - k_w \frac{T_P^0 - T_W^0}{\delta x_{WP}} \right] + \bar{S} \Delta x$$

$$a_P T_P = a_W [\theta T_W + (1 - \theta) T_W^0] + a_E [\theta T_E + (1 - \theta) T_E^0] + [a_P^0 - (1 - \theta) a_W - (1 - \theta) a_E] T_P^0 + b$$

a_P	a_P^0	a_W	a_E	b
$\theta(a_W + a_E) + a_P^0$	$\rho C_P \frac{\Delta x}{\Delta t}$	$\frac{k_w}{\delta x_{WP}}$	$\frac{k_e}{\delta x_{PE}}$	$\bar{S} \Delta x$

Explicit scheme ($\theta = 0$)

Linearize source term:

$$b = S_u + S_P T_P^0$$

All temperatures are evaluated at the old time:

$$a_P T_P = a_W T_W^0 + a_E T_E^0 + [a_P^0 - (a_W + a_E - S_P)] T_P^0 + S_u$$

a_P	a_P^0	a_W	a_E
a_P^0	$\rho C_P \frac{\Delta x}{\Delta t}$	$\frac{k_w}{\delta x_{WP}}$	$\frac{k_e}{\delta x_{PE}}$

Assessment explicit scheme

$$a_P T_P = a_W T_W^0 + a_E T_E^0 + [a_P^0 - (a_W + a_E - S_P)] T_P^0 + S_u$$

- Backward differencing scheme $O(1)$
- All coefficients should be positive:

$$a_P^0 - a_W - a_E > 0$$

In case of uniform grid and constant k :

$$a_P^0 > a_W + a_E \quad \Rightarrow \quad \rho C_P \frac{\Delta x}{\Delta t} > \frac{2k}{\Delta x} \quad \Rightarrow$$

$$\Delta t < \rho C_P \frac{(\Delta x)^2}{2k}$$

- Efficient for simple conduction calculations

Fully implicit scheme ($\theta = 1$)

Linearize source term:

$$b = S_u + S_P T_P$$

All temperatures are evaluated at the new time:

$$a_P T_P = a_W T_W + a_E T_E + a_P^0 T_P^0 + S_u$$

a_P	a_P^0	a_W	a_E
$a_P^0 + a_W + a_E - S_P$	$\rho C_P \frac{\Delta x}{\Delta t}$	$\frac{k_W}{\delta x_{WP}}$	$\frac{k_e}{\delta x_{PE}}$

Assessment Implicit scheme

$$a_P T_P = a_W T_W + a_E T_E + a_P^0 T_P^0 + S_u$$

- Forward differencing scheme $O(1)$
- Stability guaranteed: all coefficients are positive
- Accuracy ensured by choosing small Δt :

$$N_{CFL} = \frac{\Delta t |u_{\max}|}{\Delta x_{\min}} < 1$$

Courant-Friedrichs-Lewy stability condition: Courant number ≤ 1

- Recommended for general-purpose transient calculations

Crank-Nicolson scheme ($\theta = 1/2$)

Linearize source term:

$$b = S_u + \frac{1}{2}S_P T_P + \frac{1}{2}S_P T_P^0$$

All temperatures are evaluated at both the old time and the new time:

$$a_P T_P = a_W \left[\frac{T_W + T_W^0}{2} \right] + a_E \left[\frac{T_E + T_E^0}{2} \right] + \left[a_P^0 - \frac{a_W}{2} - \frac{a_E}{2} + \frac{S_P}{2} \right] T_P^0 + S_u$$

a_P	a_P^0	a_W	a_E
$\frac{1}{2}(a_W + a_E) + a_P^0 - \frac{1}{2}S_P$	$\rho C_P \frac{\Delta x}{\Delta t}$	$\frac{k_w}{\delta x_{WP}}$	$\frac{k_e}{\delta x_{PE}}$

Assessment Crank-Nicolson scheme

$$a_P T_P = a_W \left[\frac{T_W + T_W^0}{2} \right] + a_E \left[\frac{T_E + T_E^0}{2} \right] + \left[a_P^0 - \frac{a_W}{2} - \frac{a_E}{2} + \frac{S_P}{2} \right] T_P^0 + S_u$$

- Central differencing scheme $O(2)$
- All coefficients should be positive:

$$a_P^0 - \frac{a_W}{2} - \frac{a_E}{2} > 0$$

In case of uniform grid and constant k :

$$a_P^0 > \frac{a_W + a_E}{2} \Rightarrow \rho C_P \frac{\Delta x}{\Delta t} > \frac{k}{\Delta x} \Rightarrow \boxed{\Delta t < \rho C_P \frac{(\Delta x)^2}{k}}$$

- Usually used in conjunction with spatial central differencing

Steady vs. Unsteady

For pure-diffusion problems:

Steady

$$a_P T_P = a_W T_W + a_E T_E + S_u$$

a_P	a_W	a_E
$a_W + a_E - S_P$	$\frac{k_w}{\delta x_{WP}}$	$\frac{k_e}{\delta x_{PE}}$

Unsteady

$$a_P T_P = a_W T_W + a_E T_E + a_P^0 T_P^0 + S_u$$

a_P	a_P^0	a_W	a_E
$a_P^0 + a_W + a_E - S_P$	$\rho C_P \frac{\Delta x}{\Delta t}$	$\frac{k_w}{\delta x_{WP}}$	$\frac{k_e}{\delta x_{PE}}$

Convection-diffusion equation

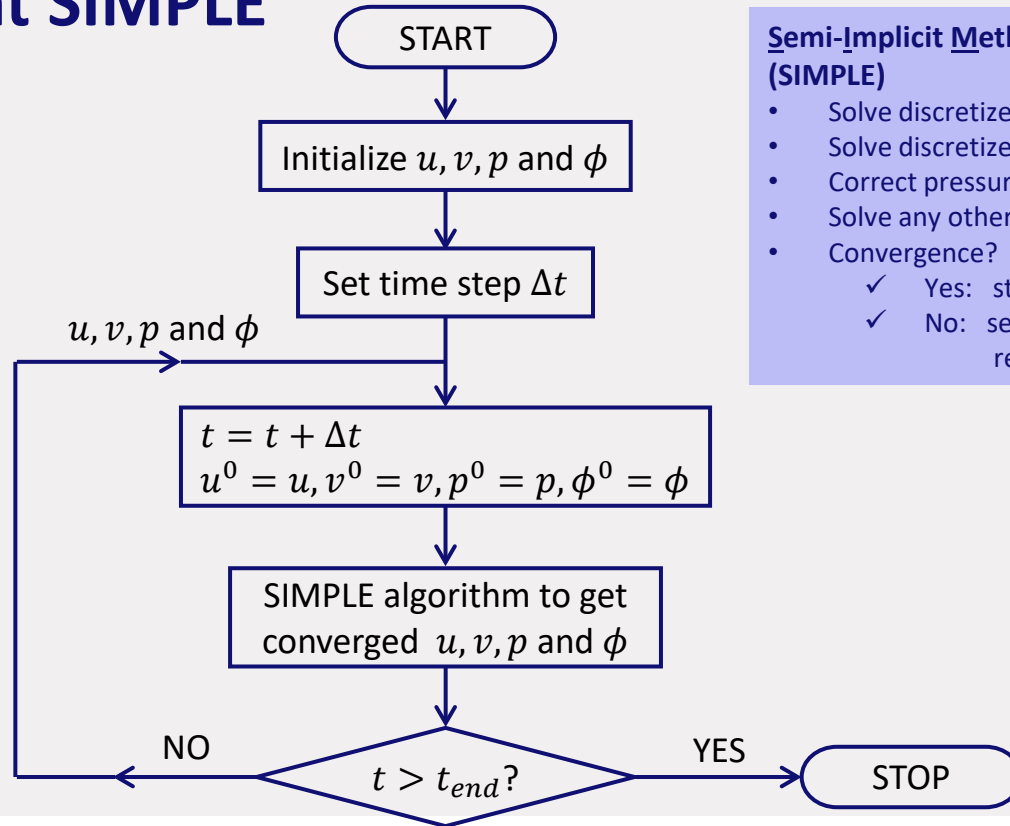
- Unsteady convection-diffusion equation:

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- Discretized form (implicit):

<i>steady</i>	<i>transient contribution</i>	<i>unsteady</i>
$a_P \phi_P = \sum a_{nb} \phi_{nb} + S_u$	$a_P^0 = \frac{\rho_P^0 \Delta V}{\Delta t}$	$a_P \phi_P = \sum a_{nb} \phi_{nb} + a_P^0 \phi_P^0 + S_u$
$a_P = \sum a_{nb} + \Delta F - S_P$		$a_P = \sum a_{nb} + a_P^0 + \Delta F - S_P$

Transient SIMPLE



Semi-Implicit Method for Pressure-Linked Equations (SIMPLE)

- Solve discretized equations for u^* and v^*
- Solve discretized equation for p'
- Correct pressure and velocities
- Solve any other discretized equations for ϕ
- Convergence?
 - ✓ Yes: stop
 - ✓ No: set $p^* = p, u^* = u, v^* = v, \phi^* = \phi$ repeat sequence

Pseudo transient simulations

- Under-relaxed x -momentum equation:

$$\frac{a_P}{\alpha_u} u_P = \sum a_{nb} u_{nb} + (p_w - p_e) A_P + b + \left[(1 - \alpha_u) \frac{a_P}{\alpha_u} \right] u_P^{n-1}$$

Previous iteration

- Transient x -momentum equation:

$$\left(a_P + \frac{\rho_P^0 \Delta V}{\Delta t} \right) u_P = \sum a_{nb} u_{nb} + (p_w - p_e) A_P + b + \frac{\rho_P^0 \Delta V}{\Delta t} u_P^0$$

Previous time step

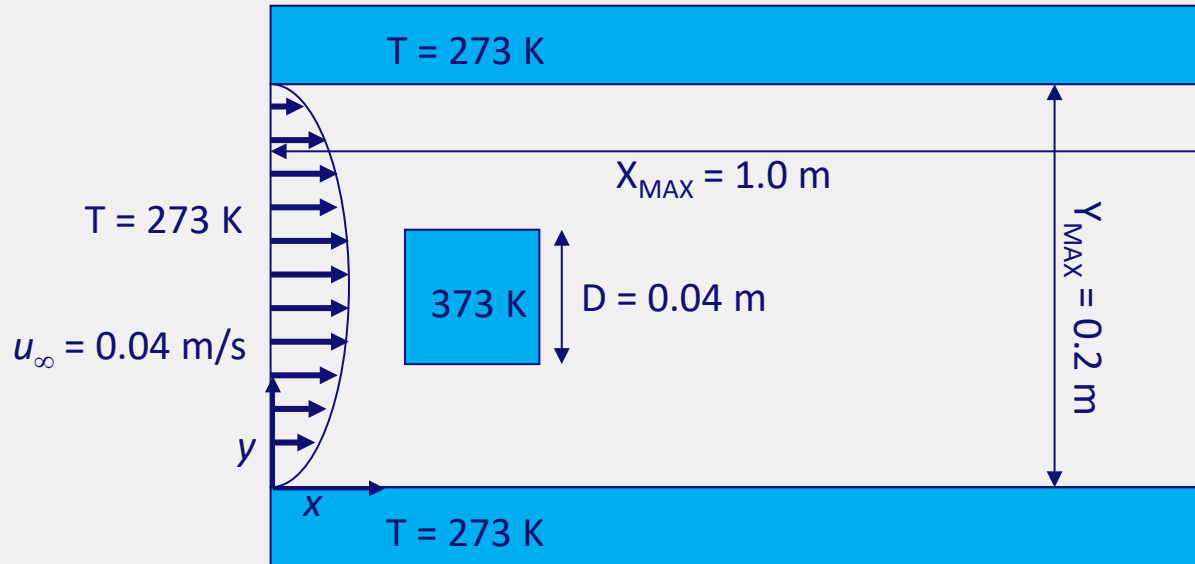
- Pseudo transient simulations:

(stabilizing iteration with faster convergence for buoyant flows, compressible flows with shocks)

$$(1 - \alpha_u) \frac{a_P}{\alpha_u} = \frac{\rho_P^0 \Delta V}{\Delta t}$$

Example: vortex shedding

$$\text{Re} = \frac{\rho u_{\infty} D}{\mu} = \frac{1.0 \cdot 0.04 \cdot 0.04}{2 \cdot 10^{-5}} = 80$$



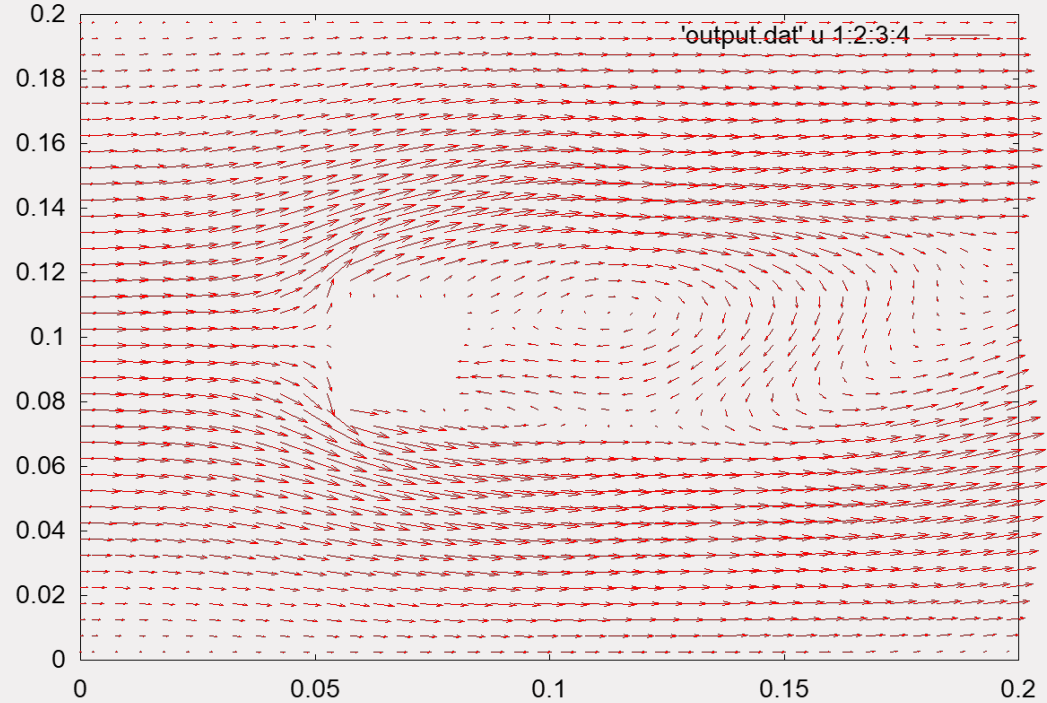
Example: vortex shedding

- **Grid: 200×40 cells**
- **$\Delta x = \Delta y = 0.005$ m**

$$\text{Re} = \frac{\rho u_{\infty} D}{\mu} = \frac{1.0 \cdot 0.04 \cdot 0.04}{2 \cdot 10^{-5}} = 80$$

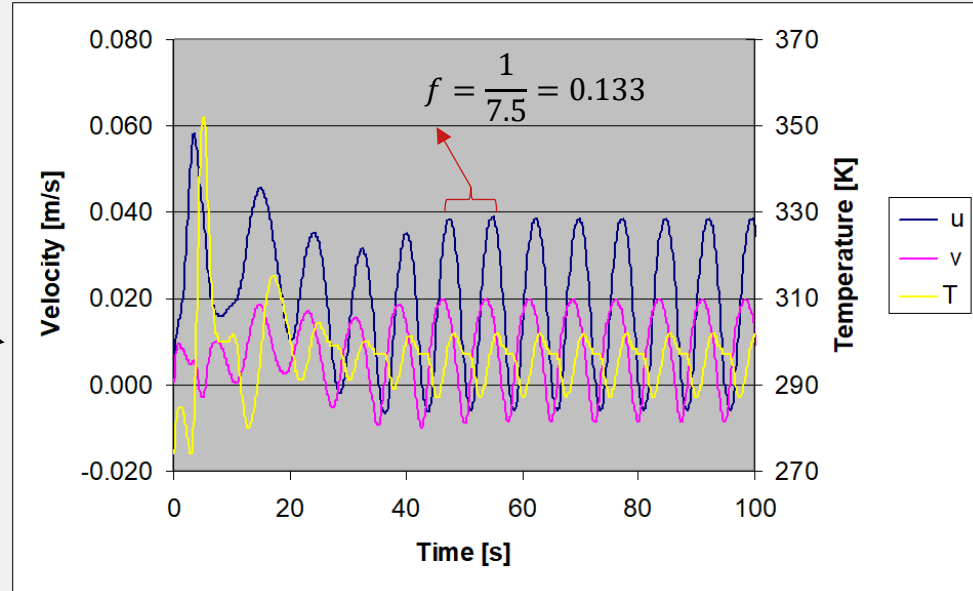
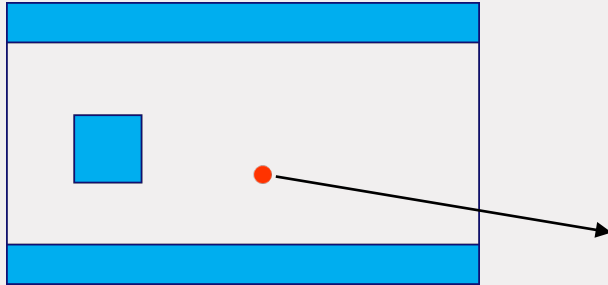
$$N_{CFL} = \frac{\Delta t |u_{\max}|}{\Delta x_{\min}} = \frac{0.1 \cdot 0.04}{0.005} = 0.8$$

Flow field

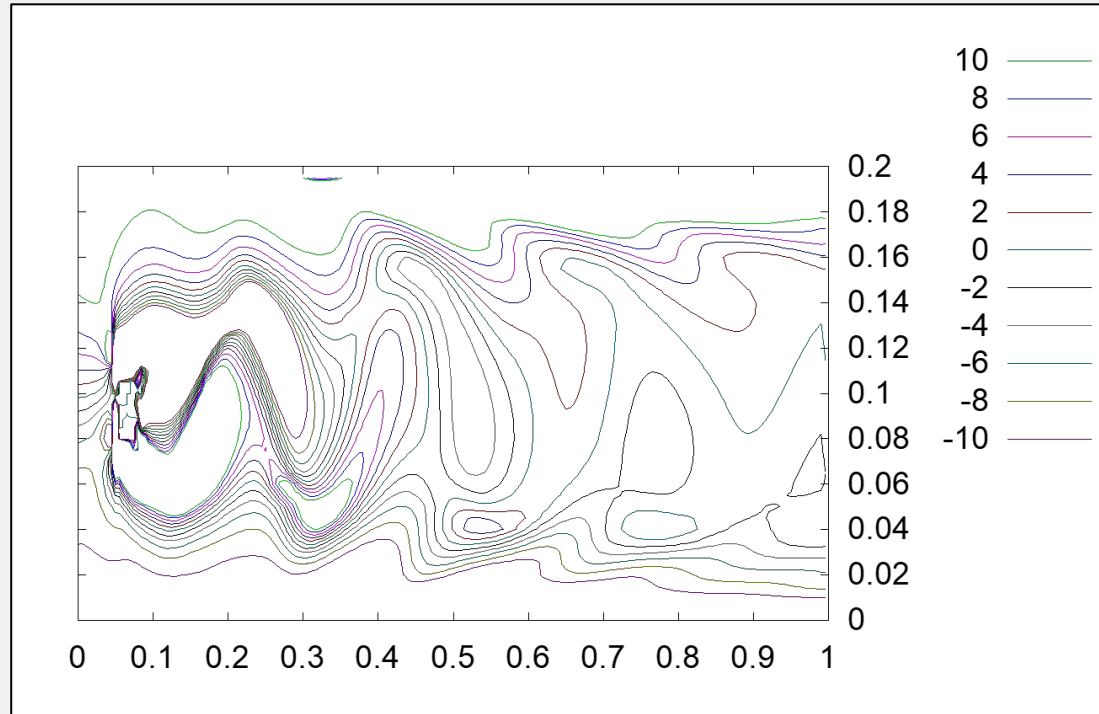


Results: vortex shedding

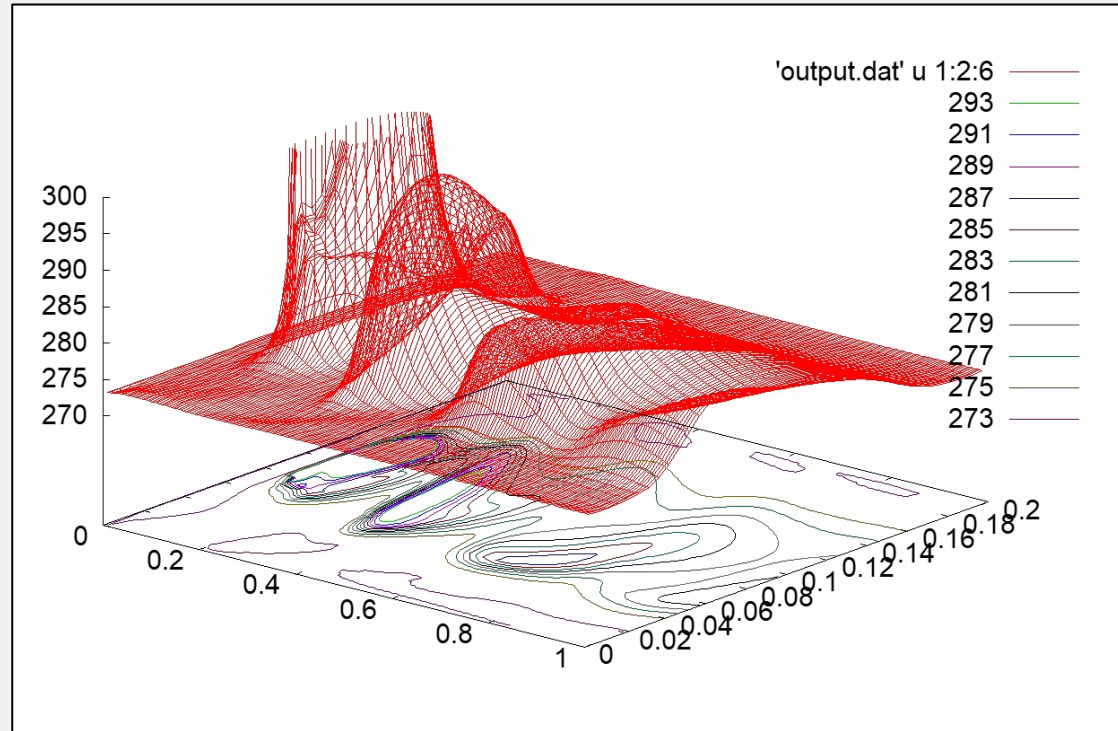
$$St = \frac{fD}{u_{\infty}} = \frac{0.04 \cdot 0.133}{0.04} = 0.133$$



Results: streamlines



Results: temperature



Experiments: vortex shedding

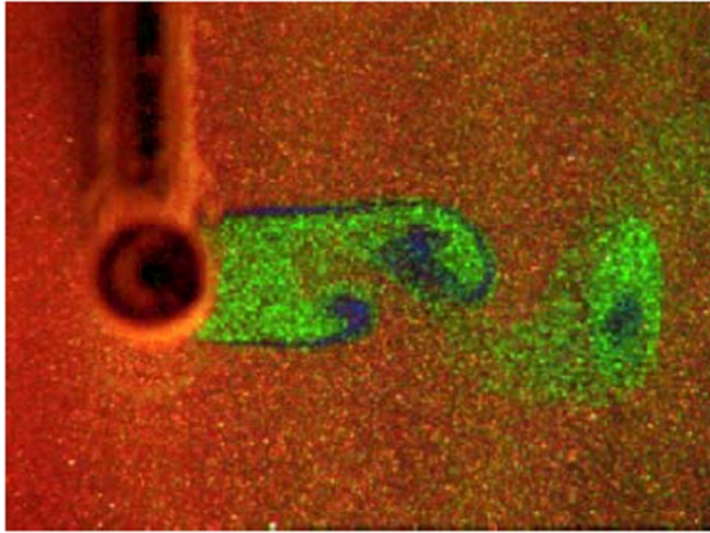


Fig. 12. Instantaneous view of the flow behind a heated circular cylinder

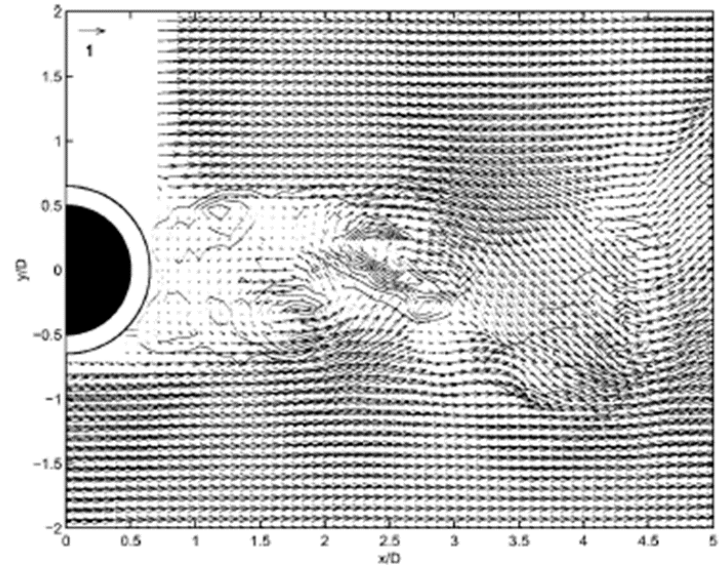


Fig. 13. Velocity and temperature fields computed using DPIV/T from image in Fig. 12 (temperature, $(T - T_\infty)/(T_c - T_\infty) > 0.01$, contour increment 0.01; velocity, u/U_∞)

Wrap up

- Unsteady convection-diffusion equation:

$$\frac{\partial(\rho\phi)}{\partial t} + \text{div}(\rho\phi\mathbf{u}) = \text{div}(\Gamma \text{grad } \phi) + S_\phi$$

- Explicit scheme $O(1)$, Crank-Nicolson scheme $O(2)$, Implicit scheme $O(1)$
- Transient SIMPLE
- Pseudo transient simulations

Solution - Convergence errors / Residuals

Numerical solution of $a_P \phi_P = \sum a_{nb} \phi_{nb} + b$ requires an iterative process (SIMPLE)

- Error (residual) for control volume i after iteration k :

$$(R_i^\phi)^k = \left| \sum a_{nb} \phi_{nb} + b - a_P \phi_P \right|_i^k$$

- Total error:

$$(\hat{R}^\phi)^k = \sum_{i=1}^N (R_i^\phi)^k$$

- Maximum error:

$$(\hat{R}_{\max}^\phi)^k = \max (R_i^\phi)^k$$

- Normalized error:

$$(\hat{R}_N^\phi)^k = (\hat{R}^\phi)^k / \sum_{i=1}^N |(a_P \phi_P)_i^k|$$

see book for different choices

- Convergence criteria when solving:

```
SMAXneeded = 1E-8; % maximum accepted error in mass balance [kg/s]
SAVGneeded = 1E-9; % maximum accepted average error in mass balance [kg/s]
while (iter <= MAX_ITER && SMAX > SMAXneeded && SAVG > SAVGneeded) % check convergence
...
```

Solution - Convergence errors / Residuals

- Decreasing residuals means the solution may not yet be converged
- Increasing residuals is normally a bad sign
- Correct residuals only mean that the solution has been converged (*convergence*), not that the solution is correct (*accuracy*)
- Different variables should be monitored to judge convergence (*e.g.* forces, moment, temperature, pressure, mass balance)